

Some industry experts' suggestions

► "The primary components in solar energy systems are charge controller and inverter design, which are required to convert the solar energy to appropriate DC level and subsequently to AC. A strong hold in DC-DC converters (charge controller) and inverter (to produce AC) design comes with having both rigorous theoretical understanding in electronics circuits and hands-on laboratory work experience. My suggestion would be to buy a small solar panel and start building simple circuits to get some exposure, and once he or she gets a hang of it, then try to build more advanced circuits which have greater efficiency and reliability."

—**Ganesh Shankar, co-founder and managing director, FluxGen Engineering Technologies**

► "Working in solar industry is a gratifying experience, where every day you go home with a sense of accomplishment. My suggestion to people looking into joining solar industry would be to keep yourself apprised with solar trends and remain updated on government policies and regulations in solar as these are key to the solar industry. For engineering department, we prefer candidates who have strong design suite knowledge (Ultiboard and Multisim). There are some simple solar programs such as PVSyst and SAM, which are desirable in candidates."

—**Advait Kumar, HR division, Saurya EnerTech**

► "Indian industry needs to develop products such as solar tracking systems and grid tied inverters. So I would suggest engineers to take a leap into their concepts existing in the imported products and also develop new concepts and alternate solutions for the applications. Grid inverter may not have an alternate solution but certainly there is a simpler solution to solar tracking system. Similarly, for commercial applications, an engineer has to think in a different way to drive LEDs to save power and get more illumination. Indian engineers should make this possible."

—**P. Chow Reddy, managing director, Interleaved Technologies**

► "Three important factors to be put in to learn as much as possible on the power electronics side of solar are hard work, dedication and time. Graduates need to try to get hands-on experience and knowledge, and present themselves as an experienced solar power electronics engineer by say 2016-17 and, I believe, that person will be a highly valuable commodity. Patience is important as the solar industry in India picks up. But use that time to become a solar power electronics guru. Your life will be just fine from then on."

—**Deyan Alex, director of operations, PowerIt Renewable Energy Pvt Ltd**

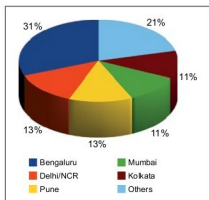


Fig. 3: Top locations where placement happens (Data courtesy: TimesJobs.com)

In terms of geographical locations, apart from Bengaluru being on the top with 31 per cent jobs, Delhi NCR and Pune offer 13 per cent each whereas Mumbai and Kolkata each offer 11 per cent of the jobs.

Opportunities for engineers

Like most electronics design engineering jobs, even in power electronics, fresh engineering graduates are best suited for testing and validation, looking at it from an organisational point

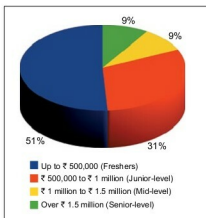


Fig. 4: Salary package (Data courtesy: TimesJobs.com)

of view as well as from an employee point of view, believes Shankar.

Shankar adds, "With the experience of testing and validation, a fresh engineering graduate is more equipped to design and construct superior-quality power electronic circuitry at a later point of time. Having said this, the fresh engineering graduate can very much participate in the design process with senior engineers, which may give fresh perspective to the design team."

Entry-level roles are mainly going to be on the R&D side of solar and power electronics.

"Most fresher jobs (masters candidates) are either in design or testing department. There is an increasing trend, at least in overseas solar firms, to hire candidates as analysts with a broad job description, where freshers can get a feel of different components before specialising," informs Advait Kumar, HR Division, Saurya EnerTech.

On a different note, Reddy informs that for a fresher, the solar power plant field restricts the person's growth. One needs to grow from basics of power electronics to ultimately work on power plants. Whereas in domestic solar products, an engineer will be concentrating at the component-level working on small products, giving the design engineer a picture of how he should behave for the power plants.

Talking about an upcoming field in the solar sector, Reddy says, "The world is working on LED lighting with solar power. This is a right combination for the power-saving LEDs. Engineers need to work further to develop efficient LED driving methods. A young designer, when given a free hand, can always make miracles in the field."

A graduate or a student with true dedication and enthusiasm for engineering can come from any institute. "For us, we have no such preferences—we are open to all. We have an internship program for undergraduate engineering students who may wish to pursue a career in renewable energy in general and solar PV systems in specific. They can by all means contact us and, based on their drive, we may like to offer them an internship opportunity in our company," says Shankar.

On a similar note, Reddy says, "I very particularly see the talent, technologies he has worked in, his versatility, adaptability and ability to develop new concepts. An individual with a diploma or masters in engineering may differ in pay but, when it comes to designing the product, I treat them equal. Engineers from IITs/NITs could